

Deep inelastic scattering on polarized lithium: the physics behind the planned measurements

2026-08-17 · companion to the cos 2φ projection report and to plans/01 (physics case) · written for students and colleagues joining the program; external numbers carry over from the fetch-verified physics-case survey

A polarized lithium beam gives the EIC something no other beam provides: spin-polarized *nuclei* light enough to compute, heavy enough to have genuine nuclear structure. ⁶Li is a spin-1 nucleus that behaves in many ways like a deuteron bound to an inert α core; ⁷Li is its spin-3/2 partner, an α plus a triton. This note explains, from the DIS formalism up, the four measurements the program is built around: the tensor structure function b₁, the double-helicity-flip ("exotic glue") structure function Δ, the polarized EMC effect, and spectator-tagged DIS on the embedded deuteron.

1 · Why lithium, and what the source must deliver

The EIC baseline includes polarized electrons, protons, deuterons, and ³He. The EPIOS white paper [1] lays out the accelerator case for adding polarized ⁶Li and ⁷Li: an atomic-beam source with optical pumping at 671 nm feeding EBIS charge breeding, with source targets P_z ≥ 0.90 (vector) and P_{zz} ≥ 0.80 (tensor) at ≥ 10¹³ atoms/s. In the ring, ⁶Li is spin-dynamically deuteron-like — its small gyromagnetic anomaly (G = −0.178) means only 81 linear depolarizing resonances up to the top energy of ~138 GeV/u, compared with 2457 for ³He — so the machine techniques developed for deuterons carry over. ⁷Li (G = +1.532) reaches ~117 GeV/u with partial snakes and tune jumps [1].

Four measurements motivate the beam, and each sits on a verified literature gap: no b₁ prediction or measurement exists for any nucleus beyond the deuteron; no gluon-transversity projection exists for any EIC target; no study of α-tagged DIS on lithium exists; and no Li-beam luminosity/polarization parameter set has been published. The program's simulations exist to turn these gaps into figures of merit.

MEASUREMENT	OBSERVABLE	ELECTRON BEAM	ION SPIN	AXIS
Polarized EMC effect (⁷ Li, ⁶ Li)	A → g ₁ ^A	polarized, helicity flipped	vector P _z	longitudinal
Tensor structure function (⁶ Li)	A _{zz} → b ₁	unpolarized	tensor P _{zz}	longitudinal (or along q)
Exotic glue (⁶ Li; ⁷ Li eligible)	cos 2φ → Δ(x, Q ²)	unpolarized	tensor P _{zz}	transverse
Tagged DIS (both isotopes)	tagged versions of all three	as above	as above	as above, correlated with spectator

2 · DIS in brief

An electron with four-momentum k scatters off a nucleus of momentum P by exchanging a virtual photon q = k − k'. Per-nucleon invariants (the convention used throughout the program: energies and structure functions per nucleon, x ≤ 1):

$$Q^2 = -q^2, \quad x = \frac{Q^2}{2P \cdot q}, \quad y = \frac{P \cdot q}{P \cdot k}, \quad Q^2 = xys, \quad W^2 = M^2 + Q^2 \frac{1-x}{x}$$

x is the momentum fraction of the struck parton, y the energy transfer fraction, W the invariant mass of everything the photon produces. The unpolarized cross section is carried by two structure functions,

$$\frac{d^2\sigma}{dx dQ^2} = \frac{4\pi\alpha^2}{xQ^4} \left[\left(1 - y + \frac{y^2}{2}\right) F_2(x, Q^2) - \frac{y^2}{2} F_L(x, Q^2) \right], \quad F_1 = \frac{F_2}{2x(1+R)}$$

with R = σ_L/σ_T. In the parton model F₂ = Σ_q e²_q x[q(x)+q̄(x)]: DIS counts quarks weighted by charge squared. For a nucleus we use F₂^A/A, so rates are quoted against luminosity per nucleon. Polarization adds structure functions on top of F₁ and F₂ — which ones depends on the target spin.

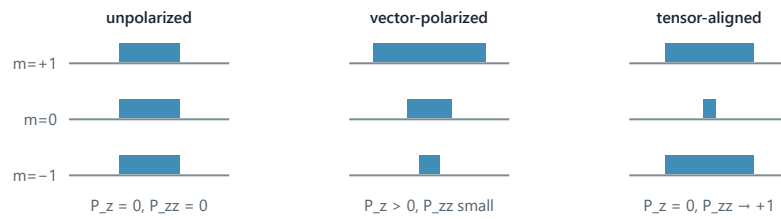
3 • What spin-1 adds

A spin-1/2 target contributes two spin structure functions, g_1 and g_2 . The workhorse is g_1 , the helicity-weighted quark distribution, measured through the double-spin asymmetry $A_{//} \approx D \cdot g_1 / F_1$ (D is a kinematic depolarization factor) with both beams longitudinally polarized and the electron helicity flipped bunch by bunch.

A spin-1 target is genuinely richer. Its spin state along a chosen axis is a set of three populations p_+ , p_0 , p_- over the projections $m = +1, 0, -1$, and two numbers characterize the fill:

$$P_z = p_+ - p_- \in [-1, 1], \quad P_{zz} = p_+ + p_- - 2p_0 = 1 - 3p_0 \in [-2, 1]$$

P_z is the familiar vector polarization. P_{zz} — the *tensor* or alignment polarization — is new: it distinguishes "spin along $\pm z$ " from "spin in the transverse plane" ($m = 0$), and it is nonzero even when equal numbers point up and down. An unpolarized beam has ($1/3, 1/3, 1/3$) and both moments vanish; a pure $m = 0$ beam has $P_z = 0$ but $P_{zz} = -2$.



Hoodbhoy, Jaffe, and Manohar showed [2] that a spin-1 target has, beyond F_1, F_2, g_1, g_2 , four tensor-polarized structure functions b_1 – b_4 , and Jaffe and Manohar identified one more, the double-helicity-flip function Δ [3]. At leading twist the tensor sector reduces to b_1 (with $b_2 = 2x b_1$ as its Callan–Gross partner) and Δ . For unpolarized electrons on a target whose spin axis makes angle θ_m with the beam, the cross section for each pure m state is [2,3]

$$\frac{d\sigma^{(m)}}{dx dy d\phi} = \frac{2y\alpha^2}{Q^2} \left[F_1 + \frac{2}{3} a_m b_1 + \frac{1-y}{xy^2} \left(F_2 + \frac{2}{3} a_m b_2 \right) - \frac{1-y}{y^2} c_m \sin^2 \theta_m \Delta(x, Q^2) \cos 2\phi \right]$$

$$a_m = \frac{1}{4} c_m (3 \cos^2 \theta_m - 1), \quad c_m = 3|\lambda_m| - 2 \rightarrow (1, -2, 1) \text{ for } m = (+1, 0, -1)$$

The weights (1, -2, 1) are the whole story of tensor observables: the $m = \pm 1$ states enter together and against $m = 0$, exactly the combination P_{zz} measures. Averaging the formula over a fill with populations p_m turns every c_m into a factor P_{zz} . Two experimental handles follow: with the spin axis *longitudinal*, the rate difference between fills gives $A_{zz} \propto b_1$; with the spin axis *transverse* ($\theta_m = 90^\circ$), the ϕ distribution of the scattered electron acquires a $\cos 2\phi$ modulation proportional to Δ . Both need tensor polarization and neither needs polarized electrons.

^7Li , with spin 3/2, has four populations and three moments (vector, tensor/quadrupole, octupole). Its rank-0 and rank-1 sectors are exact analogs of the spin-1/2 case — that is where the polarized EMC measurement lives — and its rank-2 sector is the formal analog of the spin-1 tensor sector, for which no structure-function basis has been published (an open theory item the program tracks).

4 • The four measurements

4.1 • b_1 : does a bound deuteron keep its shape?

In parton language, b_1 compares quark distributions in an aligned versus a transverse nucleus [2]:

$$b_1(x) = \frac{1}{2} \sum_q e_q^2 \left[q^{(0)}(x) - \frac{q^{(+1)}(x) + q^{(-1)}(x)}{2} \right], \quad \int_0^1 b_1(x) dx = 0 \quad (\text{Close – Kumano, no tensor sea})$$

where $q^{(m)}$ is the quark distribution in a target with spin projection m . For a collection of free spin- $1/2$ nucleons, or a pure S-wave bound state, all m states look alike and $b_1 = 0$. A nonzero b_1 is therefore a direct measure of how binding deforms the nucleon content: D-wave admixture, pion exchange, hidden-color components, or a tensor-polarized sea (which would violate the Close–Kumano sum rule [4]).

The single existing measurement is HERMES on the deuteron [5]: A_{zz} up to 2×10^{-2} over $0.01 < x < 0.45$, $b_1(x = 0.012) = 0.112 \pm 0.055 \pm 0.028$ — surprisingly large at low x , with a zero crossing near $x \approx 0.2$ and a sum-rule integral 2σ away from zero. Standard convolution calculations predict $|b_1| < 10^{-3}$ at $x \gtrsim 0.2$, an order of magnitude below the data [6]; a model with pions plus only a 0.15% hidden-color probability can reproduce it [7]. The tension is unresolved, and JLab E12-13-011 will remeasure the high- x region. The EIC adds the low- x region where HERMES saw the large signal, and the Q^2 lever arm.

How the measurement runs. The beam is delivered in fills of different tensor polarization ("tensor thirds"); with the estimator $(N_+ + N_- - 2N_0)$ over matched luminosities, all m -independent physics cancels and the statistical error on A_{zz} is $\sqrt{(2/N)/P_{zz}}$. ${}^6\text{Li}$ would be the first b_1 of any $A > 2$ nucleus — there is not even a prediction to compare to — and the ${}^6\text{Li}$ -versus-free-deuteron difference isolates what embedding in the α field does to the bound deuteron. A practical systematic tracked from the start: at collider kinematics the clean b_1 extraction prefers the spin axis along the momentum transfer $\mathbf{q}$ rather than along the beam [26].

4.2 · $\Delta(x, Q^2)$: exotic glue

The $\cos 2\phi$ term in the master formula has a special pedigree. It corresponds to the virtual photon flipping its helicity by two units between absorption and emission in the forward Compton amplitude. A spin- $1/2$ particle cannot absorb two units of angular momentum along the collinear axis — so no quark, and no bound proton or neutron, can generate Δ at leading twist. Gluons can. Δ is the structure function that the gluon transversity distribution — an interference of gluon helicity states existing only in targets with $J \geq 1$ — generates at order α_s through a photon–gluon convolution [3,9]. Any nonzero Δ in a nucleus is direct evidence for glue not inherited from individual nucleons — hence "exotic glue". It has never been measured for any target; the one prior experimental document is a JLab letter of intent [13].

The measurement: unpolarized electrons, tensor-polarized nucleus with the alignment axis in the transverse plane, and a fit of the azimuthal distribution about that axis. Averaged over the fill, the modulation amplitude and its inversion back to Δ are

$$A^{\cos 2\phi} = -\frac{1-y}{y^2} \frac{\Delta}{D_\phi}, \quad D_\phi = F_1 + \frac{1-y}{xy^2} F_2$$

The expected size spans per-mille to percent. The only absolute normalizations available are moment estimates: a bag-model calculation by Sather and Schmidt for the spin- $3/2$ Δ^{++} baryon, $\int x \Delta dx = -0.012 \alpha_s$ [9], and lattice-QCD calculations by Detmold and Shanahan of gluon transversity in the ϕ meson — clearly nonzero, at an unphysical pion mass [10] — with NPLQCD finding the analogous matrix element roughly ten times suppressed in the deuteron [11]. Percent-level asymmetries appear only in proton–deuteron Drell–Yan under the maximal assumption $\Delta_{Tg} = \Delta_g$, which its authors call optimistic [12]. No calculation exists for ${}^6\text{Li}$ or any $A > 2$ nucleus, and the literature brackets nuclear Δ in both directions — suppressed as in the lattice deuteron, or enhanced by binding like the EMC effect [1]. The projection report ports the bag moment to ${}^6\text{Li}$ as a labeled scenario, shows that one EIC year measures the amplitude at $\geq 20\sigma$ per bin, and spells out the caveats.

The coherent variant. The same modulation can be measured in coherent diffraction, $e {}^6\text{Li} \rightarrow e' X {}^6\text{Li}(\text{g.s.})$, where the nucleus stays in its ground state and recoils intact. There the $\cos 2\phi$ modulation mixes two mechanisms: the nuclear *shape* (a deformed gluon density modulates the diffractive slope — the mechanism computed for polarized deuterons in the only existing calculation of this kind [23]) and exotic glue proper. ${}^6\text{Li}$ is a sharp discriminator because its quadrupole moment is anomalously small ($Q = -0.0806 \text{ fm}^2$, about $50\times$ smaller than ${}^7\text{Li}$'s): the shape term is bounded and t-linear, so a flat-in-t modulation cannot be a shape effect. Detecting the intact recoil is its own experimental story — the nucleus keeps exactly the beam rigidity ($A/Z = 2$)

and emerges within ~2 mrad of the beam, visible only in the Roman-Pot transverse-momentum tail at IP6, or at the secondary focus of a second interaction region [25,27] — told in the companion report.

4.3 • The polarized EMC effect: g₁ of a bound proton

The EMC effect — the ~10–20% suppression of F₂ per nucleon in a nucleus at 0.3 ≲ x ≲ 0.7 — has been measured for forty years without a settled explanation. Its spin analog has never been measured at all: does the *spin* structure of a bound nucleon change more, less, or the same as its momentum structure? The observable is the polarized EMC ratio

$$\Delta R_A(x) = \frac{g_1^A(x)}{P_p g_1^p(x) + P_n g_1^n(x)}$$

where P_p, P_n are the effective nucleon polarizations from nuclear structure. ⁷Li is nearly ideal: quantum Monte Carlo gives P_p = +0.866, P_n = -0.037 [17] — to a good approximation, polarized ⁷Li is one polarized proton plus six spectators, with computable corrections. The model stakes are set by two camps: Cloët–Bentz–Thomas (mean-field, NJL) predict a polarized EMC effect about twice the unpolarized one [14]; Tronchin–Matevosyan–Thomas predict them equal [15]. A 5%-level measurement discriminates the two, and they map onto the competing mean-field versus short-range-correlation pictures of the EMC effect itself. A further EIC-only handle: the Q² evolution of g₁ is driven by the polarized gluon distribution, and the predicted *gluon-spin* EMC effect is larger still, with ⁷Li named the most promising case [16]. JLab E12-14-001 will measure ⁷Li at x > 0.06; the EIC extends to x < 0.06 and adds the Q² arm, in collider kinematics free of target-fragment absorption. For this observable both beams are polarized longitudinally and the electron helicity flips bunch-by-bunch; the error scales as 1/(P_eP_Z√N), so it uses the vector polarization the source delivers most readily.

4.4 • Tagged DIS: operating on the embedded deuteron

⁶Li ≈ α + d is more than an analogy: cluster-model calculations give the deuteron-like pair a polarization of ~0.87 inside ⁶Li, carrying over 90% of the nuclear spin [18]. If the α spectator is *detected* — it carries beam velocity into the far-forward spectrometers — the event is known to have struck the deuteron cluster, and the ⁶Li beam becomes a source of polarized deuterons at controlled virtuality. The formal machinery is spectator tagging (Cosyn–Weiss [20]): measure the spectator momentum, extrapolate in its transverse momentum to the free-nucleon (here free-cluster) pole, and recover free-particle structure with binding effects removed by construction rather than modeled. Tagged *tensor* asymmetries are predicted to reach O(1) at spectator momenta ~300 MeV/c, where the D-wave dominates — enormous compared with the inclusive asymmetries above. The benchmark for the method's power is the e+³He double-tagging study of Frišćić et al. [21]: tagging beat effective-polarization unfolding by ×2–3, and by ×10 at low x. Far-forward routing decides which isotope tags where, and it follows from magnetic rigidity alone. A beam-velocity fragment with mass-to-charge A/Z bends in the hadron optics according to R = (A/Z)_{frag}/(A/Z)_{beam}:

BEAM	TAG FRAGMENT	R	DESTINATION AT IP6
⁷ Li (A/Z = 7/3)	α	0.857	Roman Pots main window (R ∈ 0.60–0.95) — works now
⁷ Li	t	1.286	over-rigid, no coverage
⁶ Li (A/Z = 2)	α or d	1.000	beam rigidity: inside the beam envelope except the p _T tail
⁶ Li	p / n	0.50 / –	Off-Momentum Detectors / ZDC (veto handles)

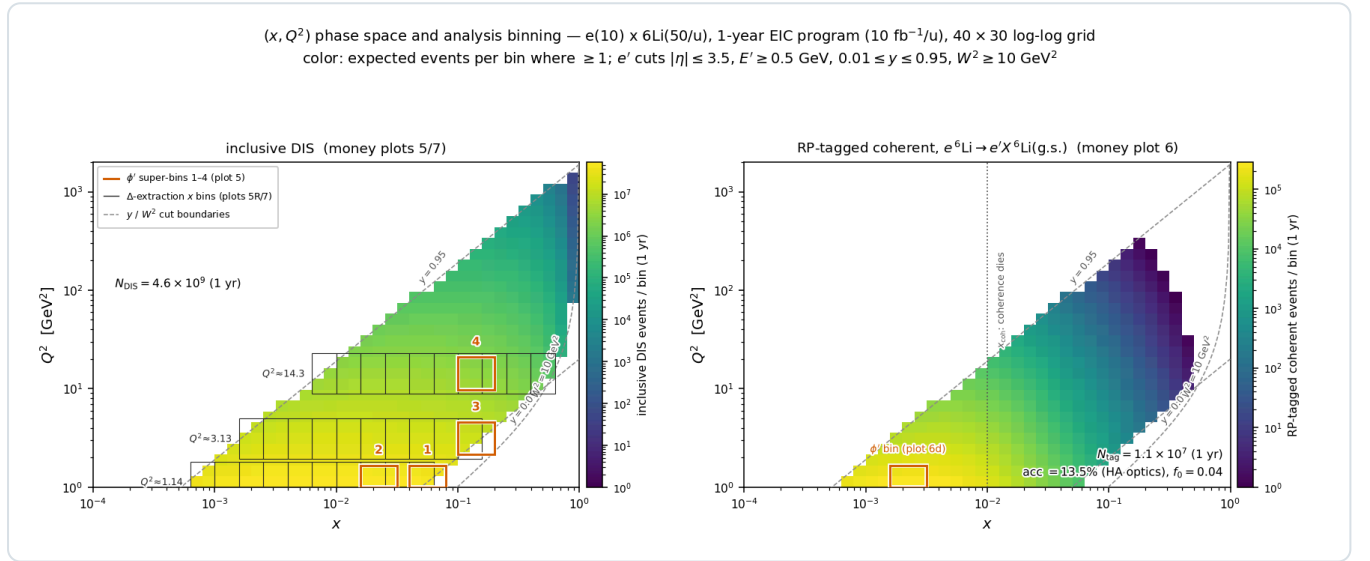
So the tagging program leads with ⁷Li at IP6, while ⁶Li α-tagging (and the coherent intact-⁶Li channel, same rigidity problem) is the physics case for the second interaction region's secondary focus [27]. This isotope asymmetry was itself a finding of the simulation program.

5 · From polarization to error bars

All projections use luminosity per nucleon, with $10 \text{ fb}^{-1}/u \approx \text{one year}$ and $100 \text{ fb}^{-1}/u \approx \text{ten years}$ at nominal EIC rates, and in-ring polarization placeholders $P_z = 0.7$, $P_{zz} = 0.6$ (ring transport from the source values is an open accelerator question). The statistical errors behind every money plot are

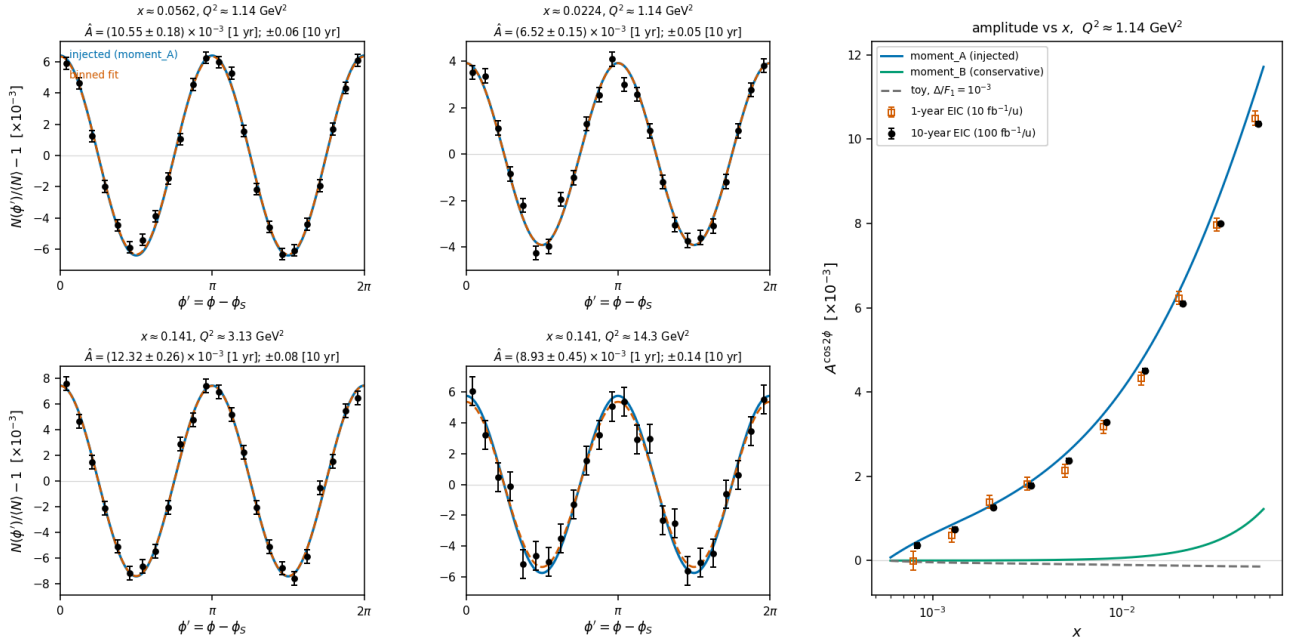
$$\delta A_{\parallel} = \frac{1}{P_e P_z \sqrt{N}}, \quad \delta A_{zz} = \frac{\sqrt{2/N}}{P_{zz}}, \quad \delta A^{\cos 2\phi} = \frac{\sqrt{2/N}}{P_{zz}}$$

The $\sqrt{2}$ in the tensor errors comes from $\langle \cos^2 2\phi \rangle = 1/2$ (or from the three-fill estimator variance); the polarization divides because the physics asymmetry is diluted by the fill. These formulas — and their breakdown under realistic ϕ acceptance holes, where a binned fit replaces the naive moment — are validated end-to-end by the polligen event generator's pseudo-experiments. Tensor observables have a structural advantage on systematics: the $\cos 2\phi$ modulation is self-normalized within a single fill, so relative-luminosity errors largely cancel; what remains is the polarimetry scale ($\delta P_{zz}/P_{zz} \approx 3\%$) and radiative corrections, which for tensor observables are uncalculated to date.



Where the events are. Expected events per (x, Q^2) bin in one EIC year at $e(10 \text{ GeV}) \times {}^6\text{Li}(50 \text{ GeV}/u)$. Left: inclusive DIS (4.6×10^9 accepted events) with the bins of the $\cos 2\phi$ analysis overlaid — statistics are never the problem; polarization, acceptance, and systematics are. Right: the coherent intact- ${}^6\text{Li}$ channel (1.1×10^7 tagged events), which lives entirely at $x \lesssim 0.03$ — coherence requires the photon to see the whole nucleus, which fails once $x \gtrsim 0.01$.

Nuclear gluonometry, transversely tensor-polarized ${}^6\text{Li}$, $e(10) \times 6\text{Li}(50/u)$, $P_{zz} = 0.60$
moment_A(variant=mid_x, c=-0.012, A=-0.2941, q2_ref=3.913, dilution=0.3333); ϕ' at 1 yr (10 fb $^{-1}/u$), amplitudes 1 & 10 yr — stat. only; bkg & tensor RC unquantified (plans/0)



What a discovery would look like. Projected $\cos 2\phi'$ pseudo-data (one EIC year, $P_{zz} = 0.6$) in the four highest-significance (x, Q^2) bins, with the injected moment-constrained Δ model and the binned fit; right, the extracted amplitude versus x against alternative model readings. The measurement distinguishes the models, not merely zero. Full assumptions and caveats: companion report.

6 • Further reading

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Educational companion to [reports/cos2phi_money_plots_report.html](#) (projections and caveats) and [plans/01_findings_physics_case.md](#) (sourced physics case). Generated by [reports/build_report.py](#); display math typeset at build time (matplotlib mathtext, STIX), figures from the evgen scripts. Projections use scenario structure functions with explicit bands.